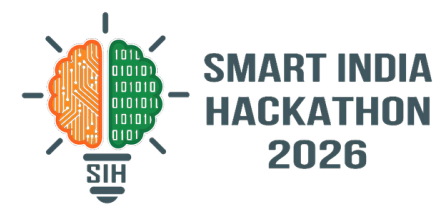
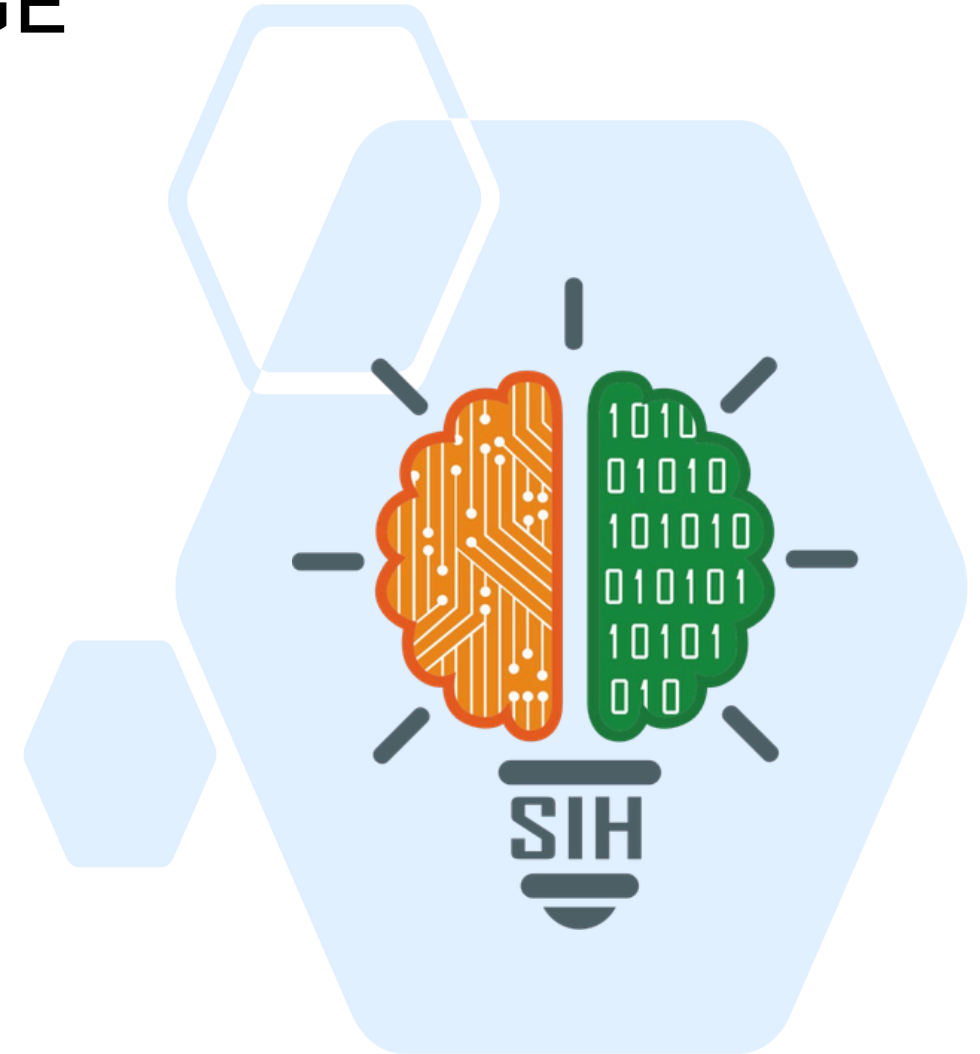


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26169
- **Problem Statement Title** – AI-Based Virtual Camera Tracking
System for Coarse Alignment of Mobile FSOC Terminals
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – —
- **Team Name** – ZeroDrift





IDEA TITLE

ZeroDrift

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

The eyes and neck of a laser-communication terminal

A physics-true virtual test-range, plus the pointing brain that finds the beacon, proves its identity and holds it — **built, measured and running today.**

ISRO · PS 26169
every figure below is measured,
not projected

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

1 SEE

A **physics-true sky** — stars, cloud, clutter and sun-glint rendered through the *same* optics as the beacon, on a real ISS orbit.



9.2 MPx/s · 30 fps · SGP4 orbit

2 IDENTIFY

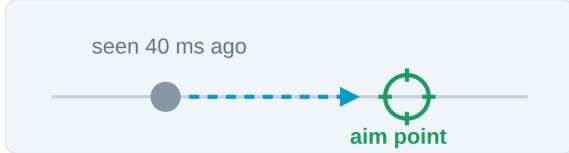
The beacon **proves who it is** by blinking at an agreed 4 Hz. A brighter star cannot fake it — it scores zero.



0 wrong locks in 64 runs

3 PREDICT

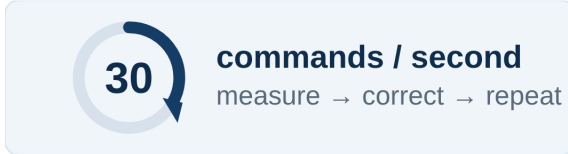
The camera only ever shows the **past**. An IMM motion model plus a Smith predictor aim where the beacon *will be*.



40 ms of mount delay cancelled

4 STEER

Pan-tilt commands go out **every single frame**; the loop re-measures and corrects, continuously.



8–21 ms of the 33 ms budget

INNOVATION AND UNIQUENESS OF THE SOLUTION

Brightness is not identity — **modulation is.**

The one hard gate every candidate must pass before the tracker will lock on to it.

Star / sun-glint

brighter, steady light



4 Hz score ≈ 0 · fails gate

REJECTED

Our beacon

dimmer, blinking at 4 Hz



4 Hz score 0.47 · passes

LOCKED

A decoy can be brighter, closer and moving — it still cannot blink on the agreed signature. That single gate is why the tracker locked a wrong target **0 times in 64 randomised runs**, and it costs no extra hardware.

HOW IT ADDRESSES THE PROBLEM



Our replay console, mid-pass on a real SGP4 ISS orbit — the same view the live demo link opens on a judge's phone. Every value is read from a logged run of the engine.

1.27s

TO ACQUIRE

98.3%

LOCK HELD

0/64

WRONG LOCKS

TECHNICAL APPROACH

CORE



Python 3.11 · NumPy · OpenCV

INTERFACE



PySide6 console · pyqtgraph live plots

ORBITS · AI



real ISS TLEs · Neural Network

QUALITY

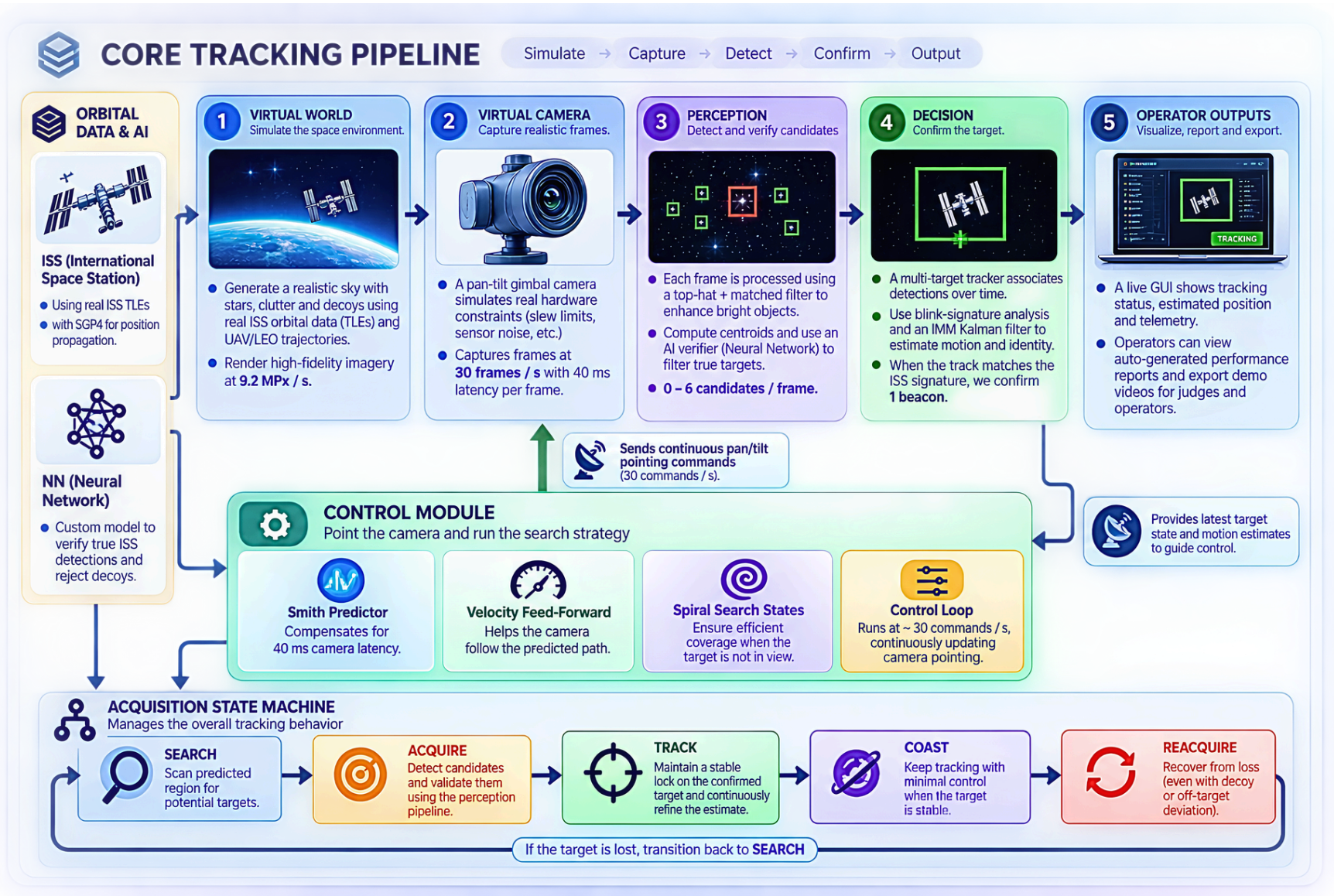


pytest suite · GitHub Actions

SHIPPING

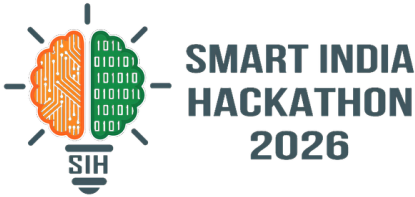


Netlify





FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

100%

73 / 73

automated tests pass on every push

100%

64 / 64

randomised runs acquired the beacon

✓

0 wrong locks

across all 64 runs — decoys never fooled it

8–21 ms of the 33 ms frame

headroom

two CPU cores — no GPU, no cloud, no dataset

not a plan — a working system, already measured

POTENTIAL CHALLENGES AND RISKS

Heavy turbulence

dims the beacon and shakes it across the frame

Bright decoys & sun-glint

confusers that look exactly like the beacon

40 ms of mount latency

high-gain loops oscillate; the mount lags the target

Sharp UAV manoeuvres

single-model filters lag hard direction changes

STRATEGIES FOR OVERCOMING THESE CHALLENGES

Coast through the fade, then re-acquire

motion model bridges dark half-cycles; spiral search restarts automatically

hard-turbulence run
1.27 s · 94.3 % lock

4 Hz blink-signature hard gate + AI verifier

a steady light scores zero; our 497-parameter net double-checks every lock

0 wrong locks / 64
AUC 0.957 vs 0.900

Two-path Smith predictor

predicts around the dead time instead of fighting it

4.2 μrad steady lag at 0.75 °/s slew

Dual-model IMM estimator

steady and manoeuvring models re-weighted every single frame

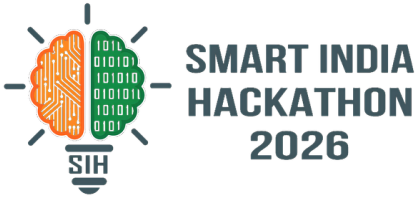
UAV-relay scenario in the standard suite

Worst case of all 64 randomised runs not the average — the worst	4.0 s slowest acquisition (median 1.27 s)	91.6 % lowest lock retention (median 97.6 %)	2.8 s a 0.53× dim beacon under 1.74× turbulence still acquired	1 command regenerates every figure — each random draw is logged
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4



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

the two-stage architecture, justified by measurement

Does the optical link actually close?

A gigabit FSOC beam is only worth anything while it stays on target. This is the fraction of a real ISS pass during which the modelled link stays closed.



Coarse alignment alone

the beam is far narrower than coarse pointing can hold

14.7 %

Coarse + modelled fine stage

ZeroDrift hands a fine steering mirror a target it can hold

99.3 %

6.8× more link time

Tracked ISS pass · 250 μrad beam · 15 cm aperture · 3.1 dB mean margin · zero outages.

Measured against ground truth the tracker never sees — not asserted.

What it replaces

An optical test bench

Cameras, gimbals, optics and lab time — procured before a single pointing algorithm can be tried, and one team can use it at a time.

₹ lakhs

ZeroDrift

A laptop you already own. Clone, run one command, and the whole develop–test–validate cycle runs — for an entire class at once.

₹ 0

- no GPU

no cloud

no dataset

no consumables

open source

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

STRATEGIC

Sovereign FSOC pointing capability for India's next-generation space and defence links — developed here, without importing an optical test bench.

ECONOMIC

A laptop replaces lakhs of rupees of hardware, and the whole develop–test–validate cycle runs at zero recurring cost.

RESEARCH & EDUCATION

Any student or ISRO lab can develop and compare PAT algorithms instantly — reproducible scenarios, honest ground-truth scoring.

SCALABLE & GREEN

One engine already covers satellite, UAV, ground and maritime links — pure software, so no consumables, no lab energy, no e-waste.

Who it serves:

ISRO / IN-SPACe optical-communication teams

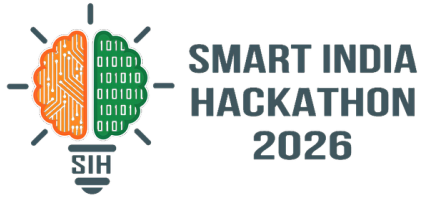
DRDO

IITs & engineering colleges

India's space-tech start-ups



RESEARCH AND REFERENCES



Research foundations

Kaymak et al., “A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,” IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)

Kaushal & Kaddoum, “Optical Communication in Space: Challenges and Mitigation Techniques,” IEEE Comm. Surveys & Tutorials, 2017. [\[link\]](#)

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management. [\[link\]](#)

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)

Smith, O.J.M., “Closer Control of Loops with Dead Time” — the Smith predictor, 1957. [\[link\]](#)

Vallado & Crawford, “SGP4 Orbit Determination” — AIAA 2008; real ISS TLE propagation. [\[link\]](#)

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation models. [\[link\]](#)



Our work —open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Live demo (replay console): zerodrift-fsoc-pat.netlify.app

Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo campaign logs · technical report · user manual · demo video.

Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods

Compared with existing approaches			
Feature	ZeroDrift	MATLAB sim	Traditional
Zero hardware cost No special equipment required	✓	✓	✗
Real-time closed loop + live console See it run. Live.	✓	✗	✓
Decoy & blink-signature discrimination Robust against spoofing	✓	✗	✗
Open source • one-command reproducible Run with a single command	✓	✗	✗